Appendix A. Performance standards (a.k.a. measurement quality objectives) for the CUPN Forest Vegetation Monitoring protocol. Standards are based on comparisons between two independent sampling events (regularly scheduled and QA/QC) conducted within a two-week span.

	Activity	Measure	Performance Standard	Reference(s)
	Plot Establishment	Record UTM X and Y coordinates of plot center	<5 m error for recorded location on geographic positioning system unit	Using the Global Positioning System, Site Selection and Plot Establishment SOPs
	Stand Data Collection	Disturbance Presence	100% agreement between observers, multiple categories (Yes/No)	Site Selection and Plot Establishment SOP
	Stand Data Collection	Tree height	±3 m averaged for plot	Canopy Closure and Stand Height Measurements SOP
	Stand Data Collection	Crown closure	± 10% averaged for plot	Canopy Closure and Stand Height Measurements SOP
	Stand Data Collection	Taxonomic Accuracy	Pseudoturnover ≤ 20% for all identified taxa and ≤ 10% for all taxa with cover class 2 or greater (i.e., estimated cover ≥ 0.1%)	Herbaceous Plant and Seedling/Sapling Measurements, Recording Species Nomenclature SOPs
	Stand Data Collection	Taxonomic Resolution	90% of observations are identified to species	Herbaceous Plant and Seedling/Sapling Measurements, Recording Species Nomenclature SOPs
	Stand Data Collection	Cover	90% of matched species within one cover class (choose from 10 classes)	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Tree Data Collection	Taxonomic Accuracy	99% of species IDs are in agreement between observers. Allowances are made within hybridizing groups (e.g., red oaks)	Tree Measurements, Recording Species Nomenclature SOPs
	Tree Data Collection	Taxonomic Resolution	95% of observations are identified to at least species level	Tree Measurements, Recording Species Nomenclature SOPs
	Tree Data Collection	Tree Mapping	Module grid coordinates (x,y) ± 1 m	Tree Measurements SOP
	Tree Data Collection	Tree diameter-at-breast-height (DBH)	±1 cm	Tree Measurements SOP
	Tree Data Collection	Tree status	100% agreement between observers (choose from 5 categories)	Tree Measurements SOP
	Tree Data Collection	Crown position	90% agreement between observers, remaining 10% within one class (choose from 5 classes)	Tree Measurements SOP
	Tree Data Collection	Vigor	80% agreement between observers, remaining 20% within one category (choose from 5 categories)	Tree Measurements SOP
	Tree Data Collection	Dieback	80% agreement between observers, remaining 20% within one category (choose from 5 categories)	Tree Measurements SOP

	Tree Data Collection	Snag decay	90% agreement between observers, remaining 10% within one class (choose from 5 classes)	Tree Measurements SOP
	Tree Data Collection	Tree Conditions	90% Agreement in recording pest/pathogen evidence on plot	Tree Measurements SOP
	Tree Data Collection	Foliage Conditions	Within one class for individual conditions (choose from 5 classes)	Tree Measurements SOP
	Tree Seedling Data Collection	Taxonomic Resolution	95% of observations are identified to at least species level	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Tree Seedling Data Collection	Taxonomic accuracy	95% of species IDs are in agreement. Allowances are made within hybridizing groups (e.g., red oaks).	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Tree Seedling Data Collection	Seedling counts	Average ± 2 individuals per corner sampled in 5-15 cm size class; ± 1 individual per corner sampled in all other size classes	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Tree Sapling Data Collection	Taxonomic resolution	95% of observations are identified to at least species level	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Tree Sapling Data Collection	Taxonomic accuracy	95% of species IDs are in agreement. Allowances are made within hybridizing groups (e.g., red oaks)	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Tree Sapling Data Collection	Tree sapling counts	Average ± 1 individual per corner sampled in all size classes	Herbaceous Plant and Seedling/Sapling Measurements SOP
	Coarse Woody Debris Data Collection	Decay class	Within one decay class between observers (choose from 5 classes)	Coarse Woody Debris SOP
	Coarse Woody Debris Data Collection	Total CWD Volume	Within 20% between observers	Coarse Woody Debris SOP
	Community Classification Data Collection	Community Classification (USNVC)	90% agreement for plots that are keyed with high confidence or fit by either crew	U. S. National Vegetation Classification Measure SOP
	Photopoint Data Collection	Image	Individual photopoint scenes capture 90% of the same area between observers	Plot Photopoints SOP